

I tested V3 on the Berkeley cases. It failed, and I want to understand why.

This is the first real outside test of my frozen Ra-226 to Ac-225 model.

PERSONAL NOTE FOR

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Professor Bernstein,

The neutron-spectrum code and papers you sent gave me something I had been missing for months: a real experiment close to the pathway I modeled. I kept my seed-42 checkpoint frozen, rebuilt the 33 and 40 MeV cases as carefully as I could, and ran the comparison without changing the model afterward.

33 MeV case Raw V3 activity was about **623 times too high.** **40 MeV case** Raw V3 activity was about **600 times too high.**

WHERE I GOT STUCK

I do not think it would be honest to tune V3 until those two points look good. Before I call this a model failure, though, I need to know whether I reconstructed the experiment correctly. The dissertation reports the recovered Ac-225 activities, while my equations predict atoms produced in the target. Those are not automatically the same quantity.

Details I can already document

- The published 33 and 40 MeV neutron-spectrum reconstruction.
- A 1 mg Ra-226 target and the reported Ac-225 activities.
- The frozen model, source hashes, inputs, and failed predictions.

Details I still cannot pin down

- Spectrum normalization or fluence at the radium target position.
- Target geometry, position, irradiation, separation, assay, and reference times.
- The exact meaning of produced versus recovered activity and any numerical Ac-227 detection limit.

What I am asking: Could you point me to where those details are defined, correct anything I misunderstood, or tell me who would know? Even a short reference would let me rebuild the comparison correctly and report the failure for the right scientific reason.

If it helps, I can send the exact inputs, reconstruction table, source code, frozen checkpoint, and the two failed outputs. I am not asking for private facility data or for Berkeley to validate my model.